Extremally projective normed spaces

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Setting. **Nor** is the category of normed spaces and bounded operators. An operator $\xi\colon X \to Y$ is *c-topologically surjective* if $c\,\xi(B_X^\circ) \supset B_Y^\circ$. For a set Λ, $\ell_1(\Lambda)$ is the Banach space of absolutely summable families and $\ell_1^0(\Lambda)$ its dense subspace of finitely supported families; the spaces $\ell_1^0(\Lambda)$ are the free objects of **Nor**. d_{BM} denotes the Banach–Mazur distance, $\overline{P}$ the completion of P.

Definition 1. A normed space P is *C-topologically projective* if for every c-topologically surjective $\xi\colon X \to Y$ between normed spaces and every bounded $\varphi\colon P \to Y$ there is a bounded $\psi\colon P \to X$ with $\xi\psi = \varphi$ and $\|\psi\| \le cC\|\varphi\|$. It is *topologically projective* if this holds for some $C \ge 1$, and *extremally projective* if it holds for every $C > 1$.

Extremal projectivity is Helemskii's extreme projectivity for normed spaces. Known before: $\ell_1^0(\Lambda)$ is extremally projective; P is extremally projective iff it is a near-retract of some $\ell_1^0(\Lambda)$; then $\overline{P} \cong \ell_1(\Gamma)$ isometrically. Helemskii (2013) and Grønbæk (2016, Conjecture 3.1) asked whether every extremally projective normed space is isometrically isomorphic to some $\ell_1^0(\Lambda)$.

Theorem 2 (dimension restrictions)**.** *Let P be topologically projective and $Z \subset P$ a subspace. Then* $\dim Z \le \max(\operatorname{dens} Z, \aleph_0)$. *Consequently P contains no infinite dimensional subspace which is complete with respect to a norm dominating the given one; in particular no infinite dimensional Banach space is topologically projective in* **Nor**.

Theorem 3 (countable dimension)**.** *Let $D \subset \ell_1$ be a dense subspace of countable algebraic dimension and $\varepsilon > 0$. There is a linear bijection T of ℓ_1 with $\|T\|\|T^{-1}\| \le 1 + \varepsilon$ and $T(D) = \ell_1^0(\mathbb{N})$. Hence, for a normed space P of countable dimension: P is extremally projective iff $\overline{P} \cong \ell_1$ or ℓ_1^n isometrically iff $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some countable Γ.*

Corollary 4 (counterexample)**.** *Let $v = (2^{-n})_{n\in\mathbb{N}}$. The space $\ell_1^0(\mathbb{N}) + \mathbb{C}v$ is extremally projective but not isometrically isomorphic to any $\ell_1^0(\Lambda)$: for $\ell_1^0(S) \subset E \subset \ell_1(S)$ the extreme points of B_E are the vectors λe_s, $|\lambda| = 1$, whose convex hull is $B_{\ell_1^0(S)}$, so an isometry $\ell_1^0(\Lambda) \to E$ forces $E = \ell_1^0(S)$.*

Example 5 (ladder space)**.** *For every limit ordinal $\alpha < \omega_1$ fix $\alpha_n \nearrow \alpha$ and put $v_\alpha = \sum_n 2^{-n} e_{\alpha_n} \in \ell_1(\omega_1)$, $P_L = \ell_1^0(\omega_1) + \operatorname{span}\{v_\alpha\}$. Then P_L is dense in $\ell_1(\omega_1)$, every countable piece $P_L \cap \ell_1(\delta)$ is extremally projective, every separable subspace of P_L has countable dimension, and P_L is not topologically projective (Fodor's lemma).*

Theorem 6 (main theorem)**.** *For a normed space P the following are equivalent:*

(i) *P is extremally projective;*

(ii) *P is topologically projective and $\overline{P} \cong \ell_1(\Gamma)$ isometrically for some set Γ;*

(iii) *$d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some set Γ;*

(iv) *for every $\varepsilon > 0$ there is Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.*

In (iii) necessarily $|\Gamma| = \dim P$, and the distance is realized by bijections $\overline{P} \to \ell_1(\Gamma)$ mapping P onto $\ell_1^0(\Gamma)$.

Thus Grønbæk's conjecture holds in the almost isometric form and fails in the isometric form. For dense subspaces of $\ell_1(\Lambda)$, topological and extremal projectivity coincide; for uncountable Λ this is a genuine restriction beyond the dimension bound.

Method. A back-and-forth construction: an automorphism of $\ell_1(\Gamma)$ of distortion $< 1 + \varepsilon$ built step by step, moving P exactly onto $\ell_1^0(\Gamma)$. Countable case: one-vector lemma (a block shift of norm $\le \delta$ swallowing one vector into $\ell_1^0(\mathbb{N})$), hitting lemma (a rank-one correction), product lemma. General case: transfinite recursion along a continuous filtration of Γ with perturbations confined to disjoint blocks, so that they add instead of multiplying; the filtration comes from closed pairs (elementary-submodel style closing off), and the blocks are handled by induction on $|\Gamma|$.

Open. An intrinsic description of the topologically projective dense subspaces of $\ell_1(\Lambda)$ for uncountable Λ.